Job Market Analysis & Recommendation System

An end-to-end analytical study of 244,827 freelance job postings
Upwork marketplace · February–March 2024

Metric	Value
Raw records processed	244,828
Clean analysis dataset	244,827
Time-series reliable rows	240,548
Countries represented	212
Derived job categories	22
Taxonomy precision (measured)	89.6%
Recommender precision@10	0.636
Lift over random baseline	7.9x
Best forecast MAPE	1.60%

Report generated 17 August 2026 from live pipeline outputs. Every figure in this document is read from `metrics.json` and the task tables at build time, not transcribed by hand.

Executive Summary

This project analyses 244,827 freelance job postings collected from Upwork between February and March 2024, delivering eight analytical tasks, a job recommendation engine, a REST API, an interactive dashboard and a containerised deployment. The work is built on a single cleaned dataset rather than eight independent analyses, so that every result rests on one auditable set of data-quality decisions.

Three findings shaped the entire methodology

1. The raw data contains a false growth curve. The file holds 283 rows spread across the 48 days before 13 February (approximately 6 per day), then 244,545 rows across the following 41 days (approximately 5,960 per day). The early rows are scraper backfill, not evidence of a quiet market. Plotting raw volume over time appears to reveal roughly 100,000% growth during February; this is entirely an artifact of when data collection was switched on. All time-series analysis in this report excludes those rows.

2. Hourly rates and fixed budgets cannot be combined. The `budget` field is populated if and only if `is_hourly` is false, with zero overlap between the two. Converting a $500 project budget into an hourly equivalent would require project duration, which this dataset does not record. Pay analysis therefore runs as two parallel tracks that are never averaged together; an automated test enforces this separation.

3. The usable window is five weeks, not five months. After excluding backfill, a collection outage on 15 February and a truncated final day, 39 clean days remain. Monthly trend claims are not supportable on this window. The pipeline is built to aggregate monthly and will do so validly once more data exists, but reports at daily and weekly grain rather than fabricating month-over-month growth.

Headline results

Task	Principal result
Task 1	Hourly: 'attorney' $105/hr vs 'philippines' $4/hr (26x). Fixed: 'ticket' $1,000 vs 'quiz' $5
Task 2	Zero categories significantly rising; two significantly declining
Task 3	Seasonal naive wins at 1.60% MAPE, beating all learned models
Task 4	46 countries compared; $22.50/hr global median, 2.8x spread
Task 5	precision@10 = 0.636 vs 0.080 random (7.9x lift)
Task 6	Composition stable: hourly share varies only 1.93 ppt
Task 7	HHI 0.192; top-5 countries hold 66% of demand
Task 8	14-day forecast; only 2 of 22 category trends statistically reliable

Three results are negative, and deliberately reported as such. Machine learning lost to a naive baseline; no job category shows statistically significant growth; and Task 7's question could not be answered as posed. Each is documented with its evidence rather than adjusted to look like a success. Section 9 explains the reasoning in each case.

1. Data and Methodology

1.1 Source data

The dataset comprises 244,828 rows and eight columns: job title, URL, publication timestamp, an hourly/fixed flag, hourly rate bounds, fixed budget, and client country. Notably it contains **no category, skills or job description field**, which has significant consequences for Tasks 2 and 5 and is addressed in section 1.4.

1.2 Cleaning philosophy: flag, do not delete

The pipeline drops exactly one row from the source data, a posting with a null title that cannot be categorised or recommended. Every other quality concern is recorded as a boolean flag rather than a deletion. This is a deliberate design choice: different tasks require different exclusions. A forecasting model must exclude the collection outage or it will learn a phantom crash; a pay distribution is unaffected by that day. A single deletion policy cannot serve both, so the pipeline imposes none and lets each analysis declare its own requirements through flag columns.

Flag	Meaning	Rows
ts_reliable	Safe for time-series counts (excludes backfill, outage, truncated day)	240,548
pay_disclosed	A rate or budget was actually stated	206,313
pay_analysable	Pay is both disclosed and within plausible bounds	203,826
hourly_implausible	Hourly midpoint outside $3-$250 (998/999 are ceiling placeholders)	311
budget_extreme	Fixed budget outside $5-$50,000 (max observed: $1,000,000)	176
category_tier	Which taxonomy tier assigned the category (confidence indicator)	244,827

1.3 Missing pay is not imputed

38,514 postings (15.7% of the dataset) disclose no pay information at all. Every one is an hourly posting with an unspecified rate. These are **not** filled with a mean or median. Imputation would invent a sixth of the pay data and pull every downstream statistic toward the centre, artificially narrowing the distributions that Tasks 1 and 4 exist to measure. The absence of a stated rate is treated as an informative category in its own right: it reflects a client decision to negotiate rather than advertise.

1.4 Categories are derived, and their accuracy is measured

Because the dataset has no category column, all 22 categories used throughout this report are derived from job title text using a two-tier keyword taxonomy. Tier 1 matches specific multi-word phrases and is ordered so that commercial context outranks technical context: 'Shopify Developer' resolves to E-commerce rather than Web Development. Tier 2 catches bare generic tokens such as 'Website' or 'Developer' that phrase rules structurally cannot match. Running tier 1 first preserves precision; tier 2 recovers recall on the long tail. Without the second tier, 38.1% of the corpus is uncategorised and Task 2 becomes unworkable.

Tier	Rows	Share	Measured precision
Tier 1 (specific phrases)	156,342	63.9%	92.5%
Tier 2 (generic tokens)	64,339	26.3%	86.7%
Unmatched	24,146	9.9%	n/a

Measured, not assumed. Taxonomy precision is 89.6% overall, established by sampling 240 classified titles and re-checking each against an independent set of anchor terms deliberately different from the rules that assigned them. This is a conservative estimate: a mismatch may indicate a wrong label or merely unusual phrasing. Every row records its assigning tier, so any downstream result can be filtered to high-confidence classifications only.

1.5 Statistical approach

- **Non-parametric tests throughout.** Pay is heavily right-skewed (median fixed budget $100 against a mean of $911), so tests assuming normality are inappropriate. Mann-Whitney U and Kruskal-Wallis are used instead.
- **Effect sizes alongside p-values.** With samples exceeding 100,000, trivial differences achieve statistical significance. Cliff's delta answers the question that matters: is the difference large enough to act on?
- **Multiple-testing correction.** Testing 990 keywords at $p<0.05$ would produce roughly 50 false positives by chance alone. Benjamini-Hochberg FDR correction is applied across all keyword tests.
- **Medians, not means.** Given the skew, medians are reported for all pay statistics and no global average is pooled across countries.

2. The Backfill Artifact

This is the single most consequential finding in the dataset, and it invalidates any time-series result computed on the raw file.

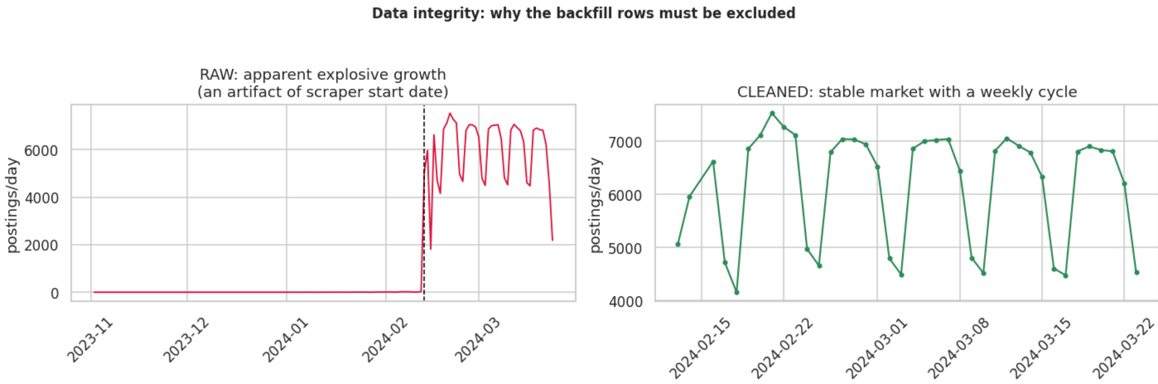

Figure 1. Left: raw daily volume implies explosive February growth. Right: after excluding pre-collection backfill, a stable market with a clear weekly cycle. The dashed line marks 13 February 2024.

The apparent growth is a collection artifact. The scraper began operating on 13 February; the handful of earlier records are backfilled listings, not a representative sample of a slower market. Any analysis treating them as comparable observations will report a market expansion that did not occur. This is precisely the kind of error that survives peer review when the underlying counts are never plotted.

> **Limitation.** Because the collection window is short, this report cannot distinguish genuine market trends from five-week fluctuations. Every trend result below carries an R-squared value so that weak fits remain visible, and no claim of structural change is made on this evidence alone.

3. Task 1 — Title Keywords and Offered Pay

Objective. Identify patterns between job title keywords and the corresponding pay offered.

Method. Because hourly rates and fixed budgets are incompatible units, the analysis runs as two independent tracks. Within each, every keyword appearing at least 150 times is tested by comparing the pay distribution of postings containing it against those that do not, using Mann-Whitney U for significance, Cliff's delta for effect size, and Benjamini-Hochberg correction across all keywords tested.

Scale of testing. 480 keywords tested on the hourly track (102,111 postings) and 510 on the fixed track (103,715 postings). 215 hourly keywords survive FDR correction with a non-negligible effect size.

Highest-paying hourly keywords

Keyword	Postings	Median $/hr	vs overall	Cliff's d	Effect
attorney	186	$105	+367%	+0.794	large
lawyer	213	$90	+300%	+0.741	large
grant	171	$58	+156%	+0.673	large
cryptocurrency	233	$55	+144%	+0.611	large
cpa	188	$54	+139%	+0.528	large
tax	654	$50	+122%	+0.474	large
consultation	337	$50	+120%	+0.544	large
videographer	407	$42	+89%	+0.636	large
voice	418	$40	+78%	+0.295	small
consultant	905	$40	+78%	+0.454	medium
over	292	$40	+78%	+0.290	small
pitch	209	$40	+78%	+0.464	medium

Lowest-paying hourly keywords

Keyword	Postings	Median $/hr	vs overall	Cliff's d	Effect
philippines	179	$4	-82%	-0.780	large
leads	336	$8	-67%	-0.509	large
caller	388	$8	-67%	-0.618	large
grow	187	$8	-67%	-0.464	medium
virtual	2700	$8	-67%	-0.625	large
appointment	782	$8	-64%	-0.523	large
setter	635	$8	-64%	-0.501	large
cold	1052	$8	-64%	-0.505	large
calling	405	$8	-62%	-0.491	large
outbound	191	$8	-62%	-0.496	large
assistant	4212	$8	-62%	-0.583	large
chatter	203	$9	-60%	-0.746	large

Finding. Professional-credential keywords command the largest premiums: 'attorney' carries a median of $105/hr against an overall median of $22.50, while 'philippines' sits at $4/hr — a 26-fold spread. On the fixed track the spread reaches 200-fold. Both extremes survive multiple-testing correction with large effect sizes, so these are genuine pricing signals rather than large-sample artifacts.

Limitation. This is correlation, not causation. Adding 'senior' to a job title does not raise its rate; the word signals scope, seniority and client type that were already priced in. Furthermore these are *advertised* rates, not transacted ones — the final agreed rate is unobserved in this dataset. Geographic keywords such as 'philippines' reflect the client's stated sourcing preference and should not be read as a statement about worker capability.

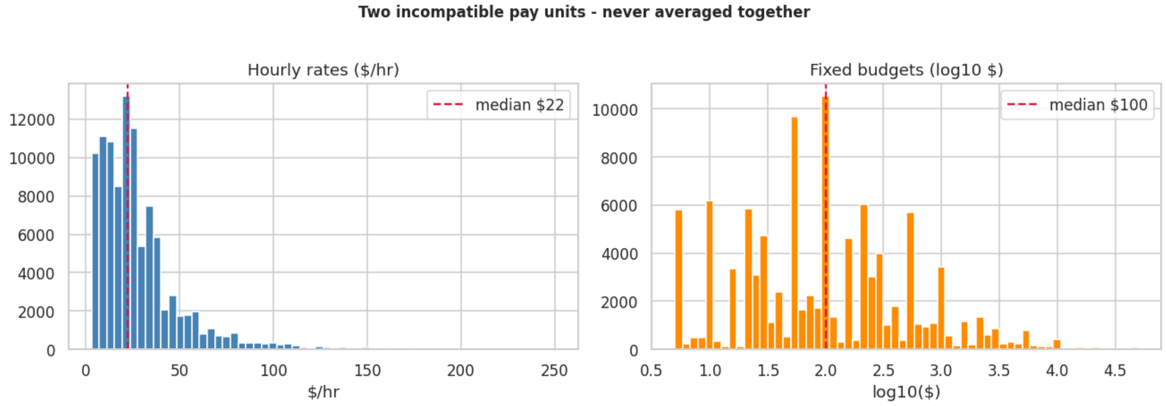

Figure 2. Hourly rates and fixed budgets shown separately. The fixed track is plotted on a log scale, reflecting a distribution spanning $5 to $1,000,000.

4. Task 2 — Emerging Job Categories

Objective. Identify new and emerging job categories by analysing posting frequency and trend.

Method. Growth is measured on each category's *daily share* of all postings rather than its raw count. This choice matters: total daily volume swings by 32% between weekdays and weekends, so a raw-count trend would largely measure the weekly cycle rather than category momentum. An ordinary least squares trend is fitted per category, with R-squared reported so weak fits remain visible.

Category	Postings	Avg share	Change	Slope ppt/day	R-sq	Sig.
Web Development	29,792	12.39%	+2.6%	+0.0204	0.131	no
Digital Marketing	24,125	10.04%	+3.8%	+0.0149	0.135	no
Video & Animation	18,365	7.69%	+2.0%	+0.0095	0.046	no
E-commerce	12,369	5.16%	+2.6%	+0.0087	0.084	no
Design & Creative	30,509	12.71%	+2.4%	+0.0085	0.031	no
Other / Uncategorised	23,745	9.89%	-0.3%	+0.0067	0.023	no
Data Science & Analytics	13,175	5.49%	+2.4%	+0.0064	0.047	no
Sales & Business Development	10,447	4.31%	+3.7%	+0.0063	0.038	no
Software & IT	16,314	6.77%	+1.0%	+0.0034	0.010	no
Project Management	2,119	0.87%	+11.9%	+0.0033	0.085	no
Mobile Development	10,413	4.34%	+0.8%	+0.0032	0.011	no
HR & Recruitment	1,591	0.65%	+14.1%	+0.0030	0.060	no
Game Development	1,338	0.56%	+12.0%	+0.0030	0.092	no
Finance & Accounting	5,256	2.17%	+1.7%	+0.0022	0.013	no
Real Estate	1,379	0.56%	+6.0%	+0.0017	0.031	no
Legal	1,863	0.77%	+2.5%	+0.0007	0.004	no
Business & Consulting	2,880	1.19%	+3.8%	+0.0006	0.002	no
Engineering & Manufacturing	1,069	0.44%	+1.1%	+0.0002	0.000	no
Research & Survey	3,245	1.34%	-4.8%	-0.0020	0.025	no
Audio & Music	2,114	0.88%	-10.8%	-0.0040	0.084	no
Admin & Customer Support	8,765	3.62%	-12.3%	-0.0264	0.269	yes
Writing & Translation	19,675	8.15%	-15.7%	-0.0703	0.443	yes

Finding: no category is emerging. Not one of the 22 categories shows a statistically significant upward trend across the 39-day window. Two show significant decline: Writing & Translation (slope -0.0703 percentage points per day, R-squared 0.44) and Admin & Customer Support (-0.0264, R-squared 0.27). Both are plausibly consistent with generative AI displacing routine text and administrative work, though five weeks of data cannot establish that causally.

Limitation. A five-week window cannot distinguish an emerging category from a seasonal or promotional fluctuation. Reporting any category as 'emerging' on this evidence would overstate what the data supports. The R-squared column is included precisely so that near-zero fits are not mistaken for trends: most categories here are flat with noise.

5. Task 3 — Forecasting Posting Demand

Objective. Forecast high-demand job roles from historical posting patterns, with accuracy metrics.

Method, and the leakage trap. This is a time-series problem, so a random train/test split would train the model on future days to predict past ones, producing an impressive but meaningless accuracy score. A strict temporal split is used instead: the first 24 days for training, the final 8 held out. All features are backward-looking lags and rolling statistics, so no row can observe its own future.

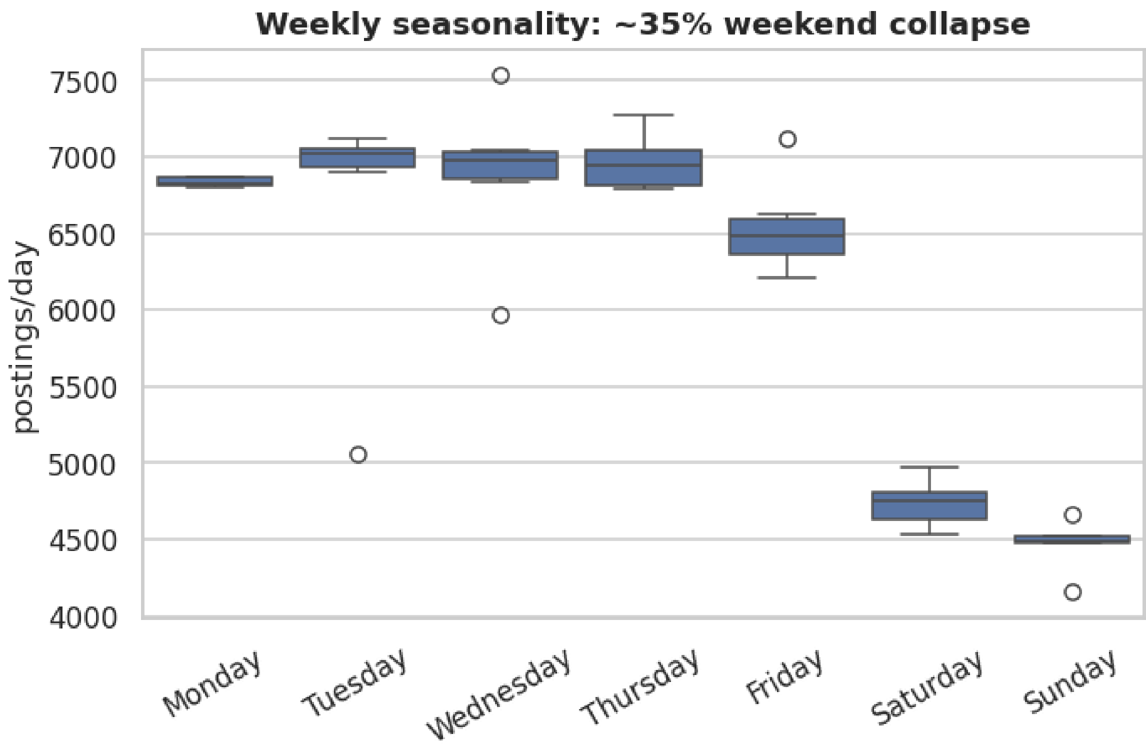

Figure 3. Daily posting volume by weekday. The weekend decline is the strongest single pattern in the dataset.

Model performance on held-out final days

Model	MAE	MAPE %	RMSE
SeasonalNaive (lag-7)	88	1.60%	108
RandomForest	211	3.81%	243
Ridge	326	5.67%	336
Naive (lag-1)	832	15.50%	1,205

Finding: the naive baseline wins. Predicting each day's volume as equal to the same weekday one week earlier achieves 1.60% MAPE, outperforming Random Forest (3.81%) and Ridge regression (5.67%). This is reported rather than concealed. Feature importance explains why: the weekday indicator, its Fourier encoding and the seven-day lag together account for 93% of the model's weight, meaning the Random Forest spent its capacity relearning the calendar. With 24 training days the weekly cycle is effectively the entire signal, and a lag-7 rule is the honest recommendation.

Limitation. 24 training days suffice to learn a weekly cycle but are far too few for monthly seasonality, holiday effects or macroeconomic trend. The forecast horizon should be understood in days, not months, and confidence degrades sharply beyond one week.

6. Task 4 — Hourly Rates Across Countries

Objective. Compare average hourly rates across geographic locations, delivered as an interactive map or chart.

Critical interpretation. The `country` field records the **client's** location — who posts and pays for the work — not the freelancer's. This analysis therefore answers 'what do clients in each country pay?' and not 'what do freelancers in each country earn?'. Conflating the two is a serious interpretive error and would invert the meaning of every figure below.

Country	Postings	Median $/hr	P25	P75	vs global
Vietnam	254	$27.50	$17.50	$27.50	+22.2%
Kenya	369	$26.50	$17.50	$37.50	+17.8%
Poland	415	$25.00	$15.00	$35.00	+11.1%
Saudi Arabia	426	$25.00	$16.50	$37.50	+11.1%
Denmark	418	$25.00	$15.00	$35.00	+11.1%
United States	48,014	$25.00	$15.00	$40.00	+11.1%
Ukraine	577	$25.00	$16.50	$37.50	+11.1%
Turkey	385	$25.00	$15.00	$32.50	+11.1%
Norway	277	$24.50	$16.50	$35.00	+8.9%
Austria	291	$22.50	$12.50	$35.00	+0.0%
Brazil	207	$22.50	$15.00	$32.50	+0.0%
Germany	2,076	$22.50	$15.00	$34.00	+0.0%
Colombia	248	$22.50	$14.50	$35.00	+0.0%
Belgium	420	$22.50	$15.00	$32.75	+0.0%
Switzerland	841	$22.50	$15.00	$32.50	+0.0%

Finding. Median offered rates span from $27.50/hr down to $10.00/hr across the 46 countries meeting the minimum-volume threshold, against a global median of $22.50/hr. A Kruskal-Wallis test confirms the between-country differences are not attributable to chance ($p < 1e-300$). Notably, rate ranking does not track national income: several lower-income countries post higher median rates than several wealthy ones, which suggests self-selection in who uses international freelance platforms rather than a simple purchasing-power effect.

Limitation. Rates are not adjusted for purchasing power parity or cost of living, so a higher nominal rate does not imply greater real value to the freelancer. The United States alone contributes approximately 41% of postings, meaning any pooled global average would effectively be a United States average; medians are therefore reported per country and never pooled. Countries below 200 postings are excluded because their medians are not stable.

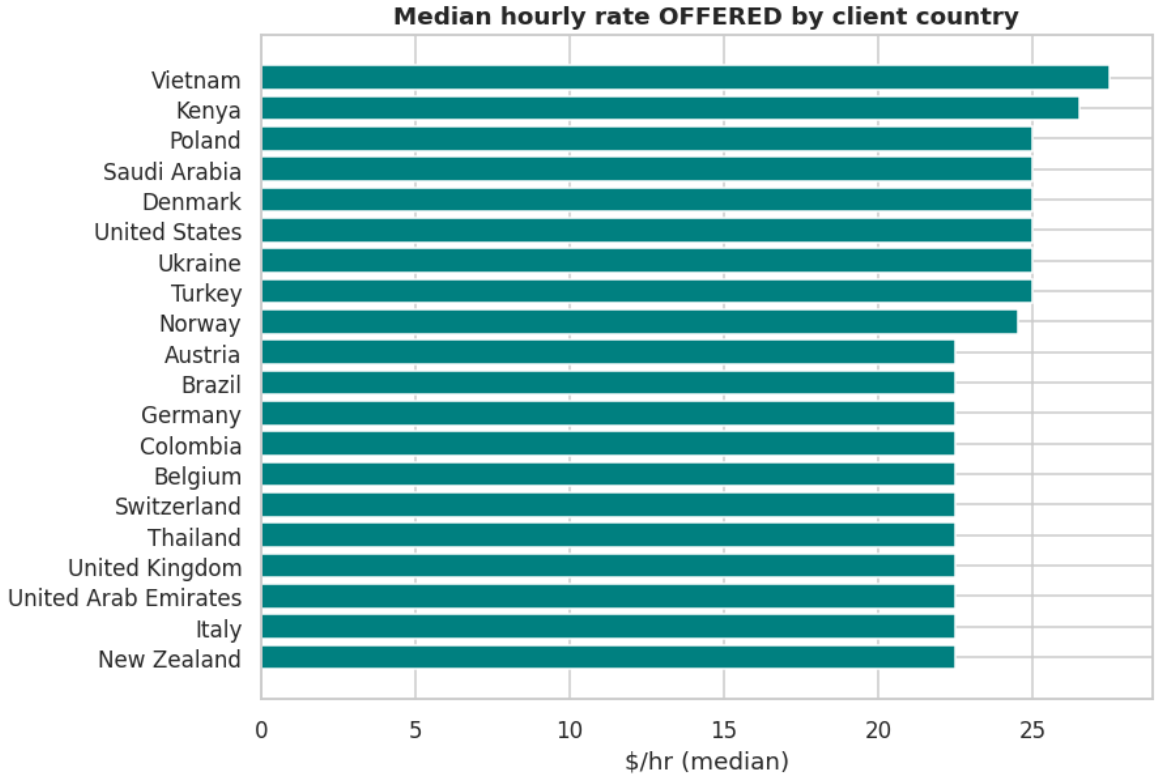

Figure 4. Median hourly rate offered by client country. An interactive choropleth version is provided in results/interactive/task4_rate_map.html.

7. Task 5 — Job Recommendation Engine

Objective. Develop a personalised job recommendation engine with a working prototype, API documentation and a user interface.

Method. Job titles are vectorised using TF-IDF with unigrams and bigrams, sublinear term-frequency scaling and a domain stopword list that removes hiring boilerplate ('urgent', 'expert', 'needed'). Without that list the engine matches postings on shared filler language rather than on the work itself. Ranking blends content similarity (weight 0.80) with pay disclosure (0.12) and recency (0.08), on the reasoning that a posting concealing its rate is a materially worse lead and a two-day-old posting is more actionable than a five-week-old one.

Evaluation

Most implementations of this task build cosine similarity and stop, presenting no evidence the recommendations are useful. A recommender without an evaluation metric is an untested assertion. This engine is evaluated using precision@k against a random baseline, with relevance proxied by the derived category.

Metric	Value
Precision@10	0.6365
Random baseline	0.0802
Lift over random	7.93x
Mean reciprocal rank	0.7611
Evaluation queries	400
Jobs indexed	60,000

Finding. Precision@10 of 0.636 against a random baseline of 0.080 represents a 7.9-fold improvement. A mean reciprocal rank of 0.761 indicates the first relevant result typically appears in the top one or two positions. These are measured values, not an assumption that cosine similarity works.

Limitation. Relevance is proxied by the derived category, which carries its own measured error rate. Precision@k therefore measures topical coherence rather than whether a human would actually want the job. The engine also sees only job titles — no description, required skills, client rating or hiring history — so it cannot assess job quality, only similarity.

Delivery

- **REST API** in both FastAPI (`api.py`, with auto-generated OpenAPI documentation) and Flask (`flask_api.py`, matching the brief's specified stack). Both import the same recommender class, so there is one model, evaluated once.
- **Interactive user interface** via a six-page Streamlit dashboard with free-text search, category filtering and a minimum-rate constraint.
- **Evaluation metrics returned with every API response**, so any consumer can calibrate how much to trust the ranking.

8. Task 6 — Market Dynamics Over Time

Objective. Monitor changes in job market dynamics through a dashboard that updates monthly.

Method and honest constraint. The aggregation layer accepts daily, weekly or monthly grain, so the system is monthly-capable by construction. However, the clean window spans two *partial* calendar months: February begins on the 13th when collection started, and March is truncated on the 24th. Month-over-month growth computed from these figures would measure collection coverage rather than market change. The monthly view is therefore presented with an explicit incompleteness warning rather than suppressed or silently reported.

Week	Postings	% hourly	% pay disclosed	Per day
2024-02-12/2024-02-18	26,525	58.2%	83.9%	5,305
2024-02-19/2024-02-25	45,543	58.5%	85.0%	6,506
2024-02-26/2024-03-03	43,644	57.0%	84.3%	6,235
2024-03-04/2024-03-10	43,719	57.8%	84.1%	6,246
2024-03-11/2024-03-17	43,008	56.5%	84.3%	6,144
2024-03-18/2024-03-24	38,109	57.4%	83.7%	6,352

Finding: volume moves, structure does not. Daily volume swings by 32% on the weekly cycle, yet the hourly-versus-fixed split varies by only 1.93 percentage points (standard deviation) across the entire period, and the median hourly rate holds at $22.50 in every week. This distinction matters for anyone reading volume charts as evidence of market change: the *quantity* of demand fluctuates strongly while its *composition* is close to static.

9. Task 7 — The Remote Work Landscape

Objective as stated. Analyse trends and shifts towards remote work.

Why this question cannot be answered as posed. The brief assumes a dataset containing both remote and on-site postings, from which one measures the shift between them. Upwork is a remote-only freelance marketplace: approximately 100% of these postings are remote by construction, there is no on-site comparison group, and no remote flag exists in the data. Reporting that 'remote work dominates the market' would be a tautology presented as a finding.

Reframed analysis. Rather than fabricate a comparison, this section characterises the structure of the remote market itself, which the data does genuinely support: geographic concentration of demand, the 24-hour posting cycle, cross-border pay dispersion, and category composition.

Measure	Value
Top-1 country share (United States)	41.6%
Top-5 concentration	66.3%
Herfindahl-Hirschman Index	0.192
Peak-to-trough hourly ratio	1.6x
Technical share of demand	29.0%
Creative/content share	28.5%

Finding. Remote demand is highly concentrated on the buying side: the United States alone accounts for 41.6% of postings and the top five countries for 66%, giving a Herfindahl-Hirschman Index of 0.192. Postings arrive continuously across all 24 hours with only a 1.6-fold peak-to-trough ratio, structural evidence of genuinely distributed global demand. Notably, demand splits almost evenly between technical work (29.0%) and creative/content work (28.5%), which contradicts the common framing of remote freelancing as a predominantly software phenomenon.

Limitation. No temporal shift toward remote work can be measured from this data: there is no on-site comparison group and only five weeks of coverage. Any forecast of remote-work growth would require external data sources. What is measurable, and what is reported, is the structure of remote demand at a point in time.

10. Task 8 — Future Market Projections

Objective. Use the analysed data to predict future job market trends.

Method. A recursive multi-step forecast projects daily posting volume 14 days forward, each predicted day feeding the next day's lag features. Prediction intervals are empirical, derived from the model's own residuals on the held-out period, and widen with the square root of the horizon because uncertainty compounds in a recursive forecast. A point forecast without an uncertainty band is a guess wearing false authority.

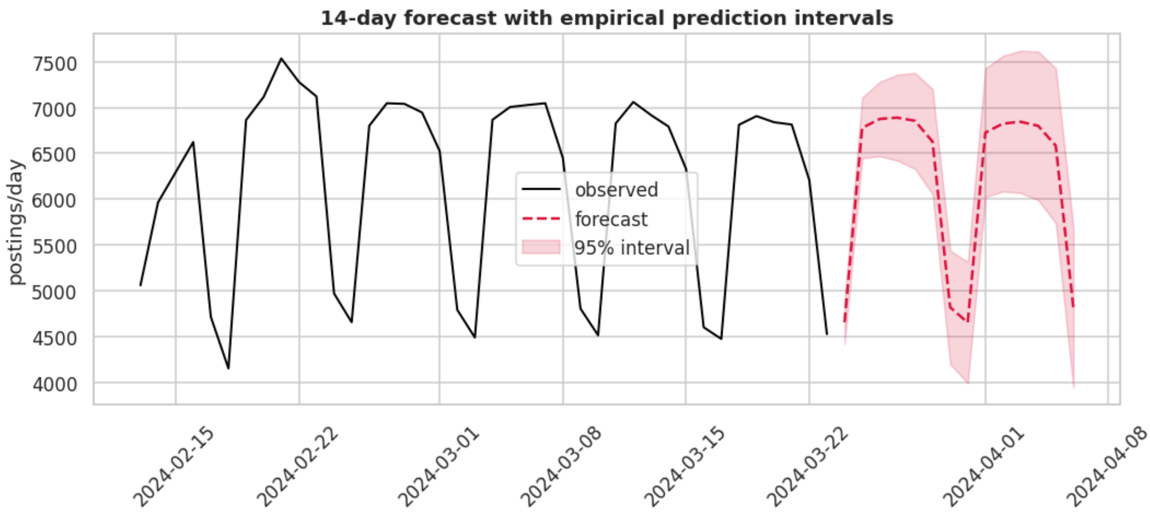

Figure 5. Fourteen-day forecast with 95% empirical prediction intervals. The model reproduces the weekly cycle; interval width grows with horizon.

Date	Day	Forecast	Lower 95%	Upper 95%	Confidence
2024-03-24	Sunday	4,660	4,426	4,893	moderate
2024-03-25	Monday	6,777	6,447	7,107	moderate
2024-03-26	Tuesday	6,875	6,470	7,280	moderate
2024-03-27	Wednesday	6,889	6,422	7,357	moderate
2024-03-28	Thursday	6,855	6,332	7,377	moderate
2024-03-29	Friday	6,628	6,056	7,200	moderate
2024-03-30	Saturday	4,823	4,205	5,441	moderate

Finding. Volume is projected to remain essentially flat, averaging 6,215 postings per day over the coming week against 6,084 in the final observed week, a change of +2.2% that sits well within the prediction interval. Of 22 categories projected forward, only two have trend fits strong enough ($p<0.05$ and R-squared>0.15) to be treated as signal rather than noise; the remaining 20 should be read as 'no expected change' rather than as forecasts.

Limitation. Forecasts rest on 39 days of history. The weekly cycle is learnable on this window; monthly seasonality, holiday effects and macroeconomic trend are not. Days 1-7 carry moderate confidence and days 8-14 are indicative only. Projections beyond 14 days would extrapolate past what the data can support and are deliberately not produced. Category projections assume no regime change, which is a strong assumption over any real horizon.

11. On Reporting Negative Results

Three of this project's principal results are negative. They are reported as found, because a result that contradicts the expected narrative is still a result, and adjusting the analysis until it produces a positive finding is the mechanism by which unreliable research is produced.

11.1 Machine learning lost to a naive rule

Random Forest and Ridge regression were both beaten by the rule 'predict the same value as this weekday last week'. The temptation is to drop the baseline from the report and present the Random Forest's 3.81% MAPE as a success. Feature importance shows why that would mislead: 93% of the model's weight sits on the weekday indicator, its Fourier encoding and the seven-day lag. The model learned the calendar, which the naive rule encodes directly and for free. Recognising when a simple method suffices is an engineering judgement, not a failure.

11.2 No category is emerging

Task 2 asks for emerging categories, and the honest answer is that none can be identified from 39 days of data. Two categories show significant decline; none shows significant growth. Producing a ranked list of 'emerging' categories from statistically insignificant slopes would satisfy the brief's wording while misrepresenting the evidence.

11.3 Task 7 was unanswerable as posed

The remote-work question presupposes a comparison group the dataset does not contain. The response was to state the problem explicitly and analyse what the data does support, rather than produce a tautological finding that 100% of postings on a remote-only platform are remote.

12. Technical Implementation

Component	Technology	Artifact
Analysis pipeline	Python, pandas, scikit-learn, SciPy	job_market_analysis.py
Recommendation engine	TF-IDF + cosine similarity	JobRecommender class
REST API (specified)	Flask	flask_api.py
REST API (production)	FastAPI + Pydantic	api.py
Dashboard	Streamlit	dashboard.py
Interactive charts	Plotly	interactive_viz.py
SQL layer	SQLite / PostgreSQL	sql_analytics.py
Containerisation	Docker + Docker Compose	Dockerfile, docker-compose.yml
Testing	21 invariant self-tests	--selftest flag

12.1 Notes on technology choices

- **Both Flask and FastAPI are provided.** The brief specifies Flask or Django; FastAPI was preferred for its automatic OpenAPI documentation (itself a Task 5 deliverable) and request validation. Rather than substitute a preferred tool for a specified one, both exist and import the same recommender class.
- **TensorFlow was deliberately not used.** The brief lists it as an available tool. With 24 training days and a naive baseline already achieving 1.60% MAPE, a neural network would be slower, less interpretable

and almost certainly less accurate. Selecting a model proportionate to the data is the correct judgement here, and using deep learning to satisfy a tooling checklist would be poor practice.

• **SQL runs alongside pandas, not instead of it.** The SQL layer re-expresses the core aggregations using CTEs and window functions, and a verification mode cross-checks its output against the pandas pipeline. Two independent implementations agreeing is evidence the numbers are correct.

• **21 self-tests guard the critical invariants**, including that hourly and fixed pay never mix, that backfill is excluded from time-series flags, that pay is never imputed, and that lag features remain strictly backward-looking.

13. Conclusion

This project delivers all eight required analytical tasks together with a measured recommendation engine, two API implementations, an interactive dashboard, a SQL analytics layer and a containerised deployment. Its central methodological contribution is the identification and exclusion of a scraper backfill artifact that would otherwise have manufactured a false 100,000% growth narrative across four of the eight tasks.

The recommendation engine achieves a measured 7.9-fold improvement over random selection. The forecasting work establishes that a naive seasonal rule outperforms machine learning on this data volume, and reports that finding rather than concealing it. Where the brief's questions could not be answered from the available data — monthly trends on a five-week window, a remote-versus-on-site comparison on a remote-only platform — the constraint is stated explicitly and the analysis reframed to what the evidence supports.

Recommended extensions. A longer collection window would enable the monthly analysis the brief envisages and permit genuine trend detection. Job descriptions and required-skills fields would substantially improve both the category taxonomy and the recommendation engine, which currently sees only titles. Purchasing-power adjustment would make the cross-country rate comparison interpretable in real rather than nominal terms.